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(54) **IMEI BINDING AND DYNAMIC IMEI PROVISIONING FOR WIRELESS DEVICES**

(52) **U.S. Cl.**
CPC **H04W 4/50** (2018.02); **H04W 8/183** (2013.01)

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(58) **Field of Classification Search**
None
See application file for complete search history.

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(56) **References Cited**

U.S. PATENT DOCUMENTS

2015/0373778 A1* 12/2015 Holtmanns H04W 88/06 455/558
2018/0060199 A1* 3/2018 Li G06F 11/2094
2022/0022014 A1 1/2022 Li et al.

* cited by examiner

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(21) Appl. No.: **18/468,661**

(57) **ABSTRACT**

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This Application sets forth techniques for binding and dynamic provisioning of international mobile equipment identifier (IMEI) values with cellular wireless service profiles, such as subscriber identity modules (SIMs) on physical SIM (pSIM) cards and electronic SIMs (eSIMs) on an embedded universal integrated circuit card (eUICC) of the mobile wireless device. When pSIMs and/or eSIMs change on the mobile wireless device, e.g., based on installation, activation, deactivation, de-installation, etc., IMEI binding logic accounts for the changes and maps IMEI values to pSIMs and/or eSIMs as required. IMEI values can be assigned based on a history of bindings between IMEI values and ICCID values of one or more eSIMs on an eUICC. A most recently used or a newly assigned IMEI value can be associated with an eSIM. Whether to assign an identical IMEI value to multiple eSIMs depends on requirements of associated cellular wireless service subscriptions.

(65) **Prior Publication Data**

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Related U.S. Application Data

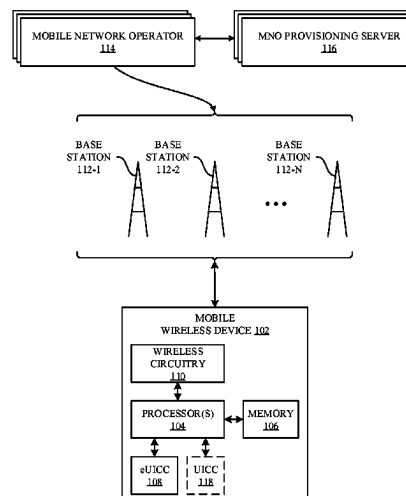
(63) Continuation of application No. 17/305,859, filed on Jul. 15, 2021, now Pat. No. 11,805,397.

(60) Provisional application No. 63/052,439, filed on Jul. 15, 2020.

(51) **Int. Cl.**
H04L 29/08 (2006.01)
H04W 4/50 (2018.01)
H04W 8/18 (2009.01)

18 Claims, 6 Drawing Sheets

100



100 →

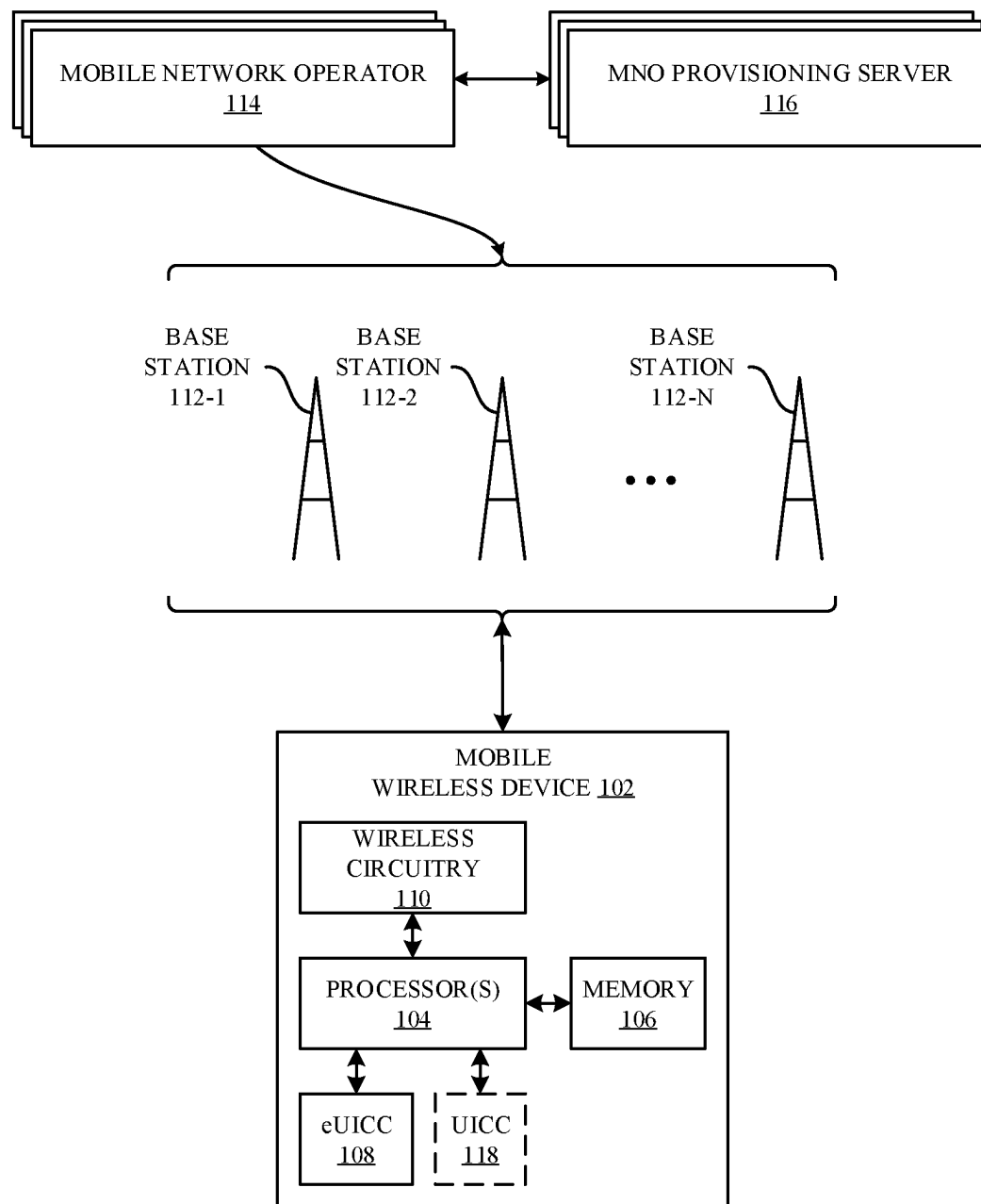


FIG. 1

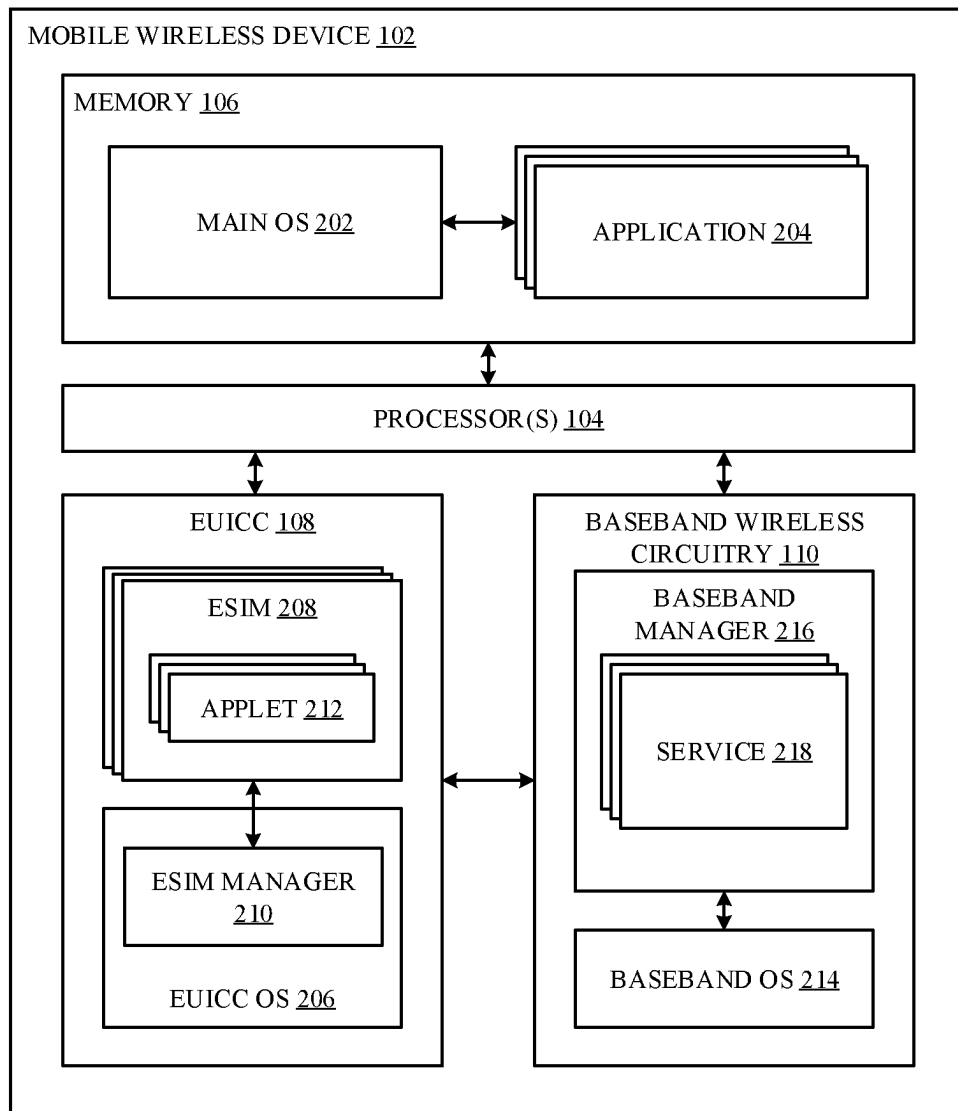
200 

FIG. 2

IMEI BINDING

300 →

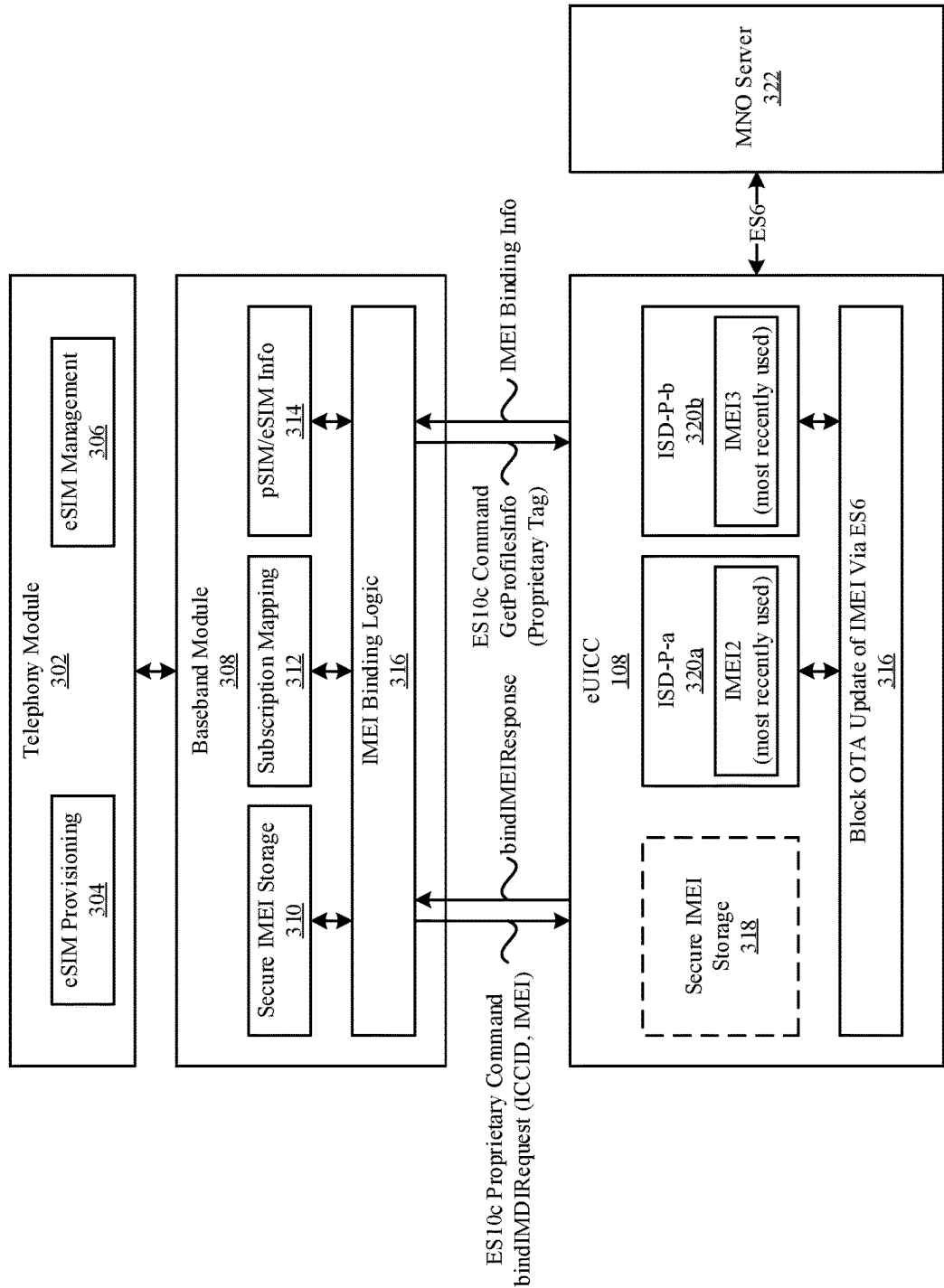


FIG. 3

400 ES10c IMEI BINDING COMMANDS

```
-- ASN1START
-- Definition of Bind IMEI
bindIMEIRequest ::= [41] SEQUENCE {
  profileIdentifier CHOICE {
    isdpAid [APPLICATION 15] OctetTo16,
    iccid ICCid
  },
  Imei [17] UTF8String (SIZE(0..64))
}
-- ASN1STOP

Response Data
The response data SHALL be coded as follows:

-- ASN1START
bindIMEIResponse ::= [41] SEQUENCE {
  bindIMEIResult INTEGER {ok(0), iccidNotFound(1), undefinedError(127)}
}
--ASN1STOP
```

FIG. 4

DYNAMIC IMEI PROVISIONING

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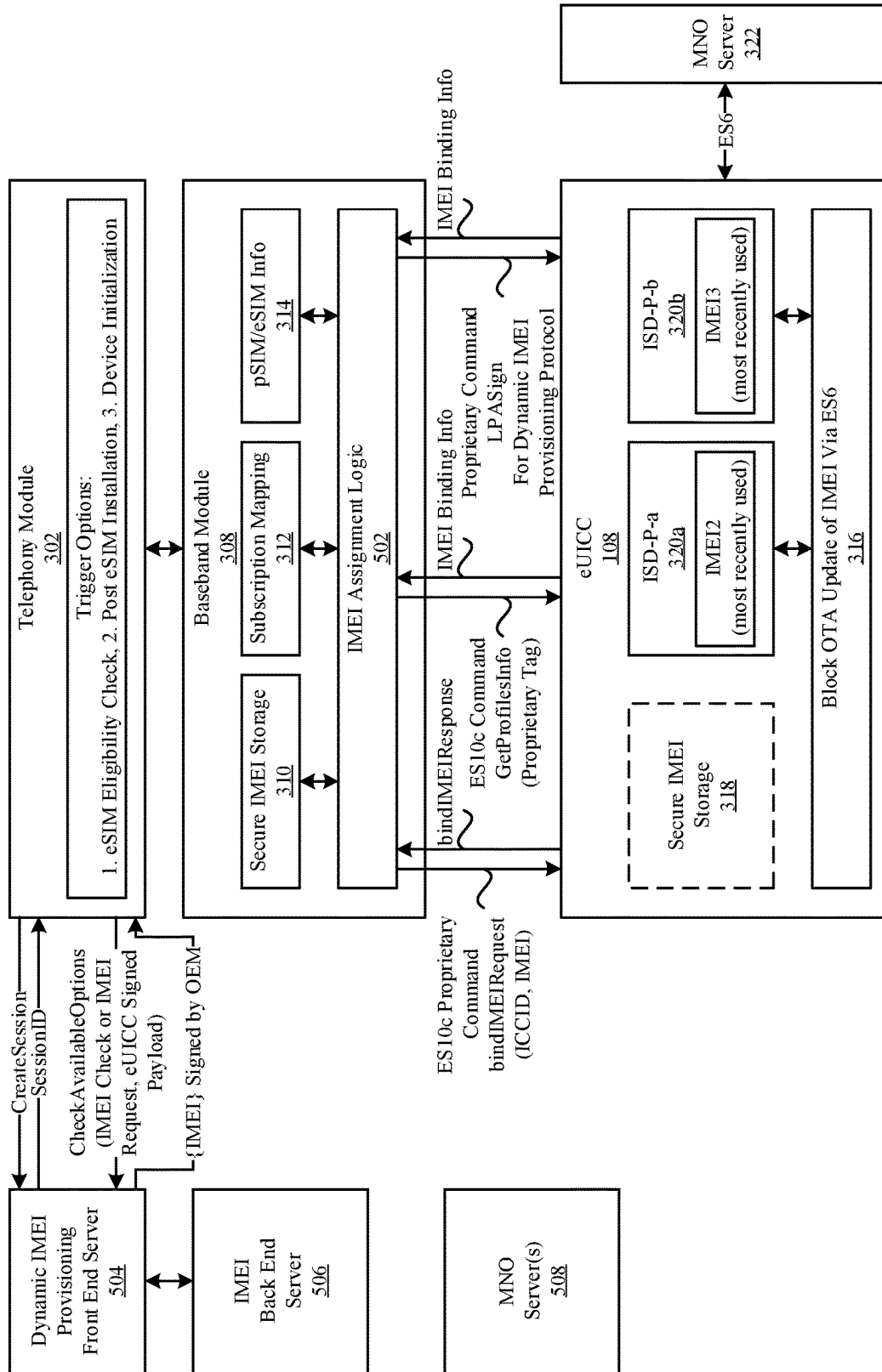
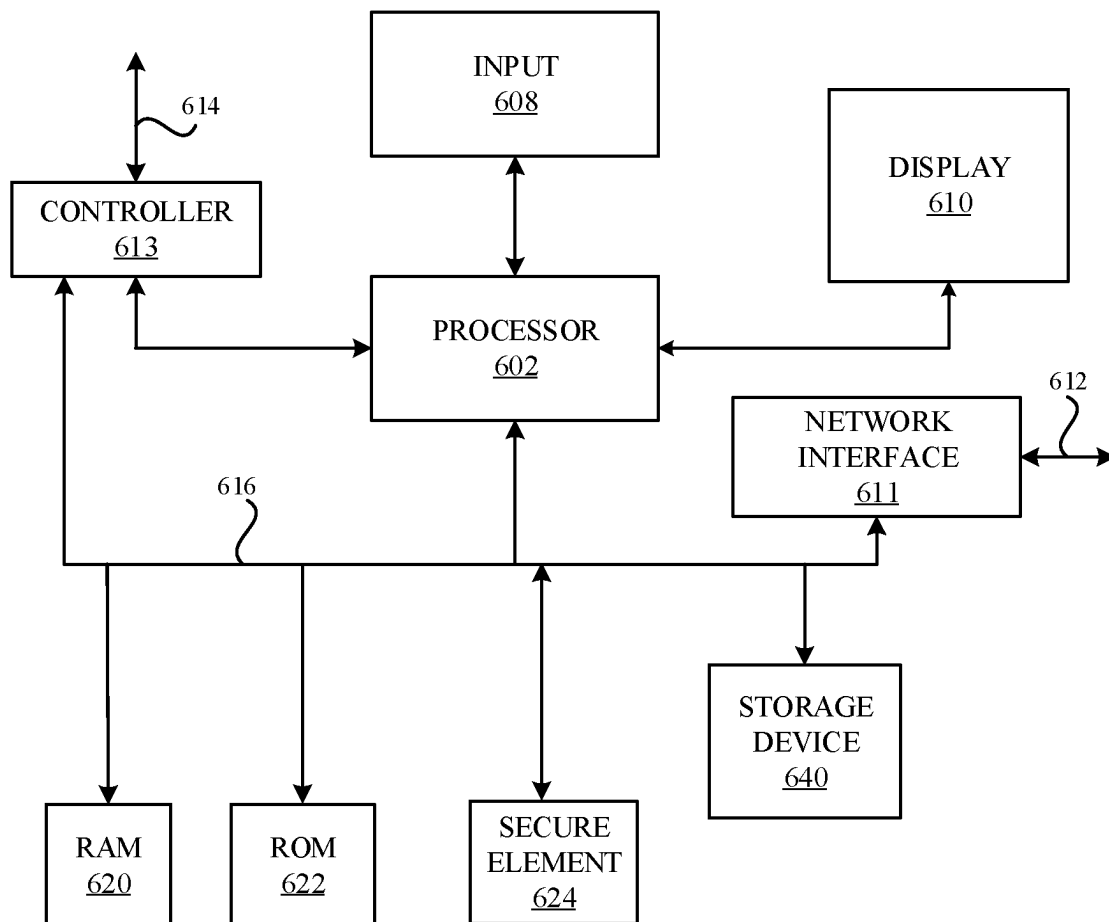


FIG. 5

600 **FIG. 6**

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IMEI BINDING AND DYNAMIC IMEI PROVISIONING FOR WIRELESS DEVICES

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. application Ser. No. 17/305,859, entitled “IMEI BINDING AND DYNAMIC IMEI PROVISIONING FOR WIRELESS DEVICES,” filed Jul. 15, 2021, which claims the benefit of U.S. Provisional Application No. 63/052,439, entitled “IMEI BINDING AND DYNAMIC IMEI PROVISIONING FOR WIRELESS DEVICES,” filed Jul. 15, 2020, the contents of all of which are incorporated by reference herein in their entirety for all purposes.

FIELD

The described embodiments set forth techniques for binding and dynamic provisioning of international mobile equipment identifier (IMEI) values for a wireless device.

BACKGROUND

Many mobile wireless devices are configured to use removable Universal Integrated Circuit Cards (UICCs) that enable the mobile wireless devices to access services provided by Mobile Network Operators (MNOs). In particular, each UICC includes at least a microprocessor and a read-only memory (ROM), where the ROM is configured to store an MNO profile that the wireless device can use to register and interact with an MNO to obtain wireless services via a cellular wireless network. A profile may also be referred to as subscriber identity module (SIM). Typically, a UICC takes the form of a small removable card, commonly referred to as a SIM card, which is inserted into a UICC-receiving bay of a mobile wireless device. In more recent implementations, UICCs are being embedded directly into system boards of wireless devices as embedded UICCs (eUICCs), which can provide advantages over traditional, removable UICCs. The eUICCs can include a rewritable memory that can facilitate installation, modification, and/or deletion of one or more electronic SIMs (eSIMs) on the eUICC, where the eSIMs can provide for new and/or different services and/or updates for accessing extended features provided by MNOs. An eUICC can store a number of MNO profiles—also referred to herein as eSIMs—and can eliminate the need to include UICC-receiving bays in wireless devices.

A hardware identifier, e.g., an international mobile equipment identifier (IMEI) value can be used to identify a wireless device with an MNO, such as for managing access to cellular wireless services. Using the same IMEI value for different cellular wireless service subscriptions may trigger fraud detection inadvertently.

SUMMARY

Representative embodiments set forth techniques for binding and dynamic provisioning of international mobile equipment identifier (IMEI) values with cellular wireless service profiles and/or with cellular wireless service subscriptions for a mobile wireless device. Examples of a cellular service profile include a subscriber identity module (SIM) on a universal integrated circuit card (UICC), also referred to as physical SIM (pSIM) on a SIM card, or an electronic SIM (eSIM) on an embedded UICC (eUICC).

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With increased flexibility in managing cellular wireless service subscriptions, such as when downloading and installing profiles (e.g., eSIMs) to access different cellular wireless services, IMEI values assigned to the mobile wireless device can be used with different profiles when the profiles are active. A mobile wireless device can include a telephony module for communicating with a baseband module that manages profile information for pSIMs and eSIMs and also accounts for different MNO cellular wireless service subscriptions, e.g., via a subscription mapping module. Multiple IMEI values can be stored on the mobile wireless device, e.g., securely in the baseband module, and mapping to various pSIMs and/or eSIMs on the mobile wireless device. As the mixture of pSIMs and/or eSIMs change on the mobile wireless device, e.g., based on installation, activation, deactivation, de-installation, etc., IMEI binding logic can account for the changes and map IMEI values to pSIMs and/or eSIMs as required. Firmware in the eUICC can ensure that IMEI values are not changed by an MNO server, e.g., during an OTA update of an eSIM. IMEI values can be assigned based on a history of bindings between IMEI values and ICCID values of one or more eSIMs on an eUICC. A most recently used IMEI value can be associated with an eSIM or a newly assigned IMEI value can be associated with the eSIM. Whether to assign the identical IMEI value to multiple eSIMs can be based, at least in part, on capabilities and/or requirements for associated cellular wireless service subscriptions and/or MNO operators thereof.

This Summary is provided merely for purposes of summarizing some example embodiments so as to provide a basic understanding of some aspects of the subject matter described herein. Accordingly, it will be appreciated that the above-described features are merely examples and should not be construed to narrow the scope or spirit of the subject matter described herein in any way. Other features, aspects, and advantages of the subject matter described herein will become apparent from the following Detailed Description, Figures, and Claims.

Other aspects and advantages of the embodiments described herein will become apparent from the following detailed description taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the described embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference numerals designate like structural elements.

FIG. 1 illustrates a block diagram of different components of an exemplary system configured to implement the various techniques described herein, according to some embodiments.

FIG. 2 illustrates a block diagram of a more detailed view of exemplary components of the system of FIG. 1, according to some embodiments.

FIG. 3 illustrates a block diagram of an exemplary system for binding international mobile equipment identifier (IMEI) values with cellular wireless service profiles and/or cellular wireless service subscriptions for a wireless device, according to some embodiments.

FIG. 4 illustrates exemplary IMEI binding commands, according to some embodiments.

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FIG. 5 illustrates a block diagram of an exemplary system for dynamic IMEI provisioning for a wireless device, according to some embodiments.

FIG. 6 illustrates a block diagram of exemplary elements of a mobile wireless device, according to some embodiments.

DETAILED DESCRIPTION

Representative applications of methods and apparatus according to the present application are described in this section. These examples are being provided solely to add context and aid in the understanding of the described embodiments. It will thus be apparent to one skilled in the art that the described embodiments may be practiced without some or all of these specific details. In other instances, well known process steps have not been described in detail in order to avoid unnecessarily obscuring the described embodiments. Other applications are possible, such that the following examples should not be taken as limiting.

In the following detailed description, references are made to the accompanying drawings, which form a part of the description and in which are shown, by way of illustration, specific embodiments in accordance with the described embodiments. Although these embodiments are described in sufficient detail to enable one skilled in the art to practice the described embodiments, it is understood that these examples are not limiting; such that other embodiments may be used, and changes may be made without departing from the spirit and scope of the described embodiments.

This Application sets forth techniques for binding and dynamic provisioning of international mobile equipment identifier (IMEI) values with cellular wireless service profiles and/or with cellular wireless service subscriptions for a mobile wireless device. Examples of a cellular service profile include a subscriber identity module (SIM) on a universal integrated circuit card (UICC), also referred to as physical SIM (pSIM) on a SIM card, or an electronic SIM (eSIM) on an embedded UICC (eUICC). Mobile wireless devices are typically provisioned with i) a single IMEI value, such as for a mobile wireless device that has a bay for receiving a single physical SIM card, or ii) with a pair of IMEI values, such as for a dual-SIM, dual-standby (DSDS) device that accommodates two physical SIM cards, or for a DSDS device that includes an eUICC (on which one or more eSIMs can be installed) and a UICC-receiving bay for a single physical SIM card. With increased flexibility in managing cellular wireless service subscriptions, such as when downloading and installing profiles (e.g., eSIMs) to access different cellular wireless services, IMEI values assigned to the mobile wireless device can be used with different profiles when the profiles are active. For mobile wireless devices that are restricted to a single physical SIM card associated with one IMEI value and only one active eSIM on an eUICC at any one time, where the active eSIM is associated with a second IMEI value, use of two IMEI values in parallel can be accommodated by assigning two distinct IMEI values during manufacture of the mobile wireless device, one IMEI value to be used with a physical SIM card and a second IMEI value to be used with an active eSIM on the eUICC of the mobile wireless device. For mobile wireless devices that allow for multiple eSIMs to be active at one time, using the same IMEI value for different eSIMs at the same time, or using the same IMEI value for a physical SIM card and an eSIM at the same time, can cause issues. In some circumstances, simultaneous use of the same IMEI value for two different SIMs/eSIMs can trigger fraud

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detection by a mobile network operator (MNO), particularly when cellular wireless services of the same MNO are accessed using the two different SIMs/eSIMs by the mobile wireless device. To address this problem, this Application proposes solutions that include mechanisms for binding IMEI values to SIMs/eSIMs and dynamic provisioning of IMEI values.

A mobile wireless device can include a telephony module for communicating with a baseband module that manages profile information for pSIMs and eSIMs and also accounts for different MNO cellular wireless service subscriptions, e.g., via a subscription mapping module. Multiple IMEI values can be stored on the mobile wireless device, e.g., securely in the baseband module, and mapping to various pSIMs and/or eSIMs on the mobile wireless device. As the mixture of pSIMs and/or eSIMs change on the mobile wireless device, e.g., based on installation, activation, deactivation, de-installation, etc., IMEI binding logic can account for the changes and map IMEI values to pSIMs and/or eSIMs as required. Firmware in the eUICC can ensure that IMEI values are not changed by an MNO server, e.g., during an OTA update of an eSIM. IMEI values can be assigned based on a history of bindings between IMEI values and ICCID values of one or more eSIMs on an eUICC. A most recently used IMEI value can be associated with an eSIM or a newly assigned IMEI value can be associated with the eSIM. Whether to assign the identical IMEI value to multiple eSIMs can be based, at least in part, on capabilities and/or requirements for associated cellular wireless service subscriptions and/or MNO operators thereof.

These and other embodiments are discussed below with reference to FIGS. 1-6; however, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes only and should not be construed as limiting.

FIG. 1 illustrates a block diagram of different components of a system 100 that is configured to implement the various techniques described herein, according to some embodiments. More specifically, FIG. 1 illustrates a high-level overview of the system 100, which, as shown, includes a mobile wireless device 102, which can also be referred to as a wireless device, a mobile device, a user equipment (UE) and the like, a group of base stations 112-1 to 112-N that are managed by different Mobile Network Operators (MNOs) 114, and a set of provisioning servers 116 that are in communication with the MNOs 114. Additional MNO infrastructure servers, such as used for account management and billing are not shown. The mobile wireless device 102 can represent a mobile computing device (e.g., an iPhone® or an iPad® by Apple®) or a cellular-capable wearable device (e.g., an Apple Watch®), the base stations 112-1 to 112-n can represent cellular wireless network entities including evolved NodeBs (eNodeBs or eNBs) and/or next generation NodeBs (gNodeBs or gNB) that are configured to communicate with the mobile wireless device 102, and the MNOs 114 can represent different wireless service providers that provide specific cellular wireless services (e.g., voice and data) to which the mobile wireless device 102 can subscribe, such as via a subscription account for a user of the mobile wireless device 102.

As shown in FIG. 1, the mobile wireless device 102 can include processing circuitry, which can include one or more processor(s) 104 and a memory 106, an embedded Universal Integrated Circuit Card (eUICC) 108, and a baseband wireless circuitry 110 used for transmission and reception of cellular wireless radio frequency signals. The baseband

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wireless circuitry **110** can include analog hardware components, such as antennas and amplifiers, as well as digital processing components, such as signal processors (and/or general/limited purpose processors) and associated memory. In some embodiments, the mobile wireless device **102** includes one or more physical UICCs **118**, also referred to as Subscriber Identity Module (SIM) cards, in addition to or substituting for the eUICC **108**. The components of the mobile wireless device **102** work together to enable the mobile wireless device **102** to provide useful features to a user of the mobile wireless device **102**, such as cellular wireless network access, non-cellular wireless network access, localized computing, location-based services, and Internet connectivity. The eUICC **108** can be configured to store multiple electronic SIMs (eSIMs) for accessing cellular wireless services provided by different MNOs **114** by connecting to their respective cellular wireless networks through base stations **112-1** to **112-N**. For example, the eUICC **108** can be configured to store and manage one or more eSIMs for one or more MNOs **114** for different subscriptions to which the mobile wireless device **102** is associated. To be able to access services provided by an MNO **114**, an eSIM is reserved for subsequent download and installation to the eUICC **108**. In some embodiments, the eUICC **108** obtains one or more eSIMs from one or more associated provisioning servers **116**. The provisioning servers **116** can be maintained by a manufacturer of the mobile wireless device **102**, the MNOs **114**, third party entities, and the like. Communication of eSIM data between an MNO provisioning server **116** and the eUICC **108** (or between the MNO provisioning server **116** and processing circuitry of the mobile wireless device **102** external to the eUICC **108**, e.g., the processor **104**) can use a secure communication channel.

FIG. 2 illustrates a block diagram of a more detailed view **200** of particular components of the mobile wireless device **102** of FIG. 1, according to some embodiments. As shown in FIG. 2, the processor(s) **104**, in conjunction with memory **106**, can implement a main operating system (OS) **202** that is configured to execute applications **204** (e.g., native OS applications and user applications). As also shown in FIG. 2, the eUICC **108** can be configured to implement an eUICC OS **206** that is configured to manage hardware resources of the eUICC **108** (e.g., a processor and a memory embedded in the eUICC **108**). The eUICC OS **206** can also be configured to manage eSIMs **208** that are stored by the eUICC **108**, e.g., by downloading, installing, deleting, enabling, disabling, modifying, or otherwise performing management of the eSIMs **208** within the eUICC **108** and providing baseband wireless circuitry **110** with access to the eSIMs **208** to provide access to wireless services for the mobile wireless device **102**. The eUICC **108** OS can include an eSIM manager **210**, which can perform management functions for various eSIMs **208**. According to the illustration shown in FIG. 2, each eSIM **208** can include a number of applets **212** that define the manner in which the eSIM **208** operates. For example, one or more of the applets **212**, when implemented in conjunction with baseband wireless circuitry **110** and the eUICC **108**, can be configured to enable the mobile wireless device **102** to communicate with an MNO **114** and provide useful features (e.g., phone calls and internet access) to a user of the mobile wireless device **102**.

As also shown in FIG. 2, the baseband wireless circuitry **110** of the mobile wireless device **102** can include a baseband OS **214** that is configured to manage hardware resources of the baseband wireless circuitry **110** (e.g., a processor, a memory, different radio components, etc.).

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According to some embodiments, the baseband wireless circuitry **110** can implement a baseband manager **216** that is configured to interface with the eUICC **108** to establish a secure channel with an MNO provisioning server **116** and obtaining information (such as eSIM data) from the MNO provisioning server **116** for purposes of managing eSIMs **208**. The baseband manager **216** can be configured to implement services **218**, which represents a collection of software modules that are instantiated by way of the various applets **212** of enabled eSIMs **208** that are included in the eUICC **108**. For example, services **218** can be configured to manage different connections between the mobile wireless device **102** and MNOs **114** according to the different eSIMs **208** that are enabled within the eUICC **108**.

FIG. 3 illustrates a diagram **300** of an exemplary system for binding international mobile equipment identifier (IMEI) values with cellular wireless service profiles (e.g., SIM cards and/or eSIMs **208**) and/or with cellular wireless service subscriptions for a mobile wireless device **102**. A telephony module **302** can be resident on a processor **104** of the mobile wireless device **102** and manage cellular wireless service profiles using an eSIM provisioning module **304** and an eSIM management module **306**. The telephony module **302** can communicate with a baseband module **308** that includes cellular wireless service profile information, such as pSIM/eSIM information **314** for one or more physical SIM cards (UICCs **118**) and/or one or more eSIMs **208** stored on the eUICC **108** of the mobile wireless device **102**. The baseband module **308** can also include a subscription mapping module **312** to account for different MNO cellular wireless service subscriptions of a user of the mobile wireless device **102** and their use with different UICCs **118** and/or eSIMs **208** installed on the eUICC **108** of the mobile wireless device **102**. The baseband module **308** can further include secure IMEI storage **310** to store IMEI values that can be associated with active pSIMs and/or active eSIMs **208** (or previously associated with inactive pSIMs and/or eSIMs **208**). An IMEI binding logic module **316** can interface with the various modules of the baseband module **308**, e.g., with secure IMEI storage **310**, subscription mapping module **312**, and/or pSIM/eSIM information **314**, and communicate between these modules and the eUICC **108**. The baseband module **308** accounts for use of the mobile wireless device **102**, including the mixture of pSIMs and eSIMs **208** that are used, and communicates with the eUICC **108**, via the IMEI binding logic **316**, to request IMEI values to associate (bind) with specific SIM/eSIM identifier values, e.g., with particular international circuit card identifier (ICCID) values. The IMEI binding logic **316** can also obtain information from the eUICC **108**, by inquiring about a present set of IMEI associations (bindings) used by the eUICC **108**. The IMEI binding logic **316** can account for changes in use of different SIMs/eSIMs to bind IMEI values to the SIMs/eSIMs in active use by the mobile wireless device **102**. Firmware in the eUICC **108** can also ensure that IMEI values used by eSIMs **208** are not changed by an MNO server **322**, e.g., by precluding over-the-air (OTA) updates to IMEI values via the ES6 interface. Blocking OTA updates of IMEI binding information by the eUICC **108** can also protect against a denial of service (DoS) attack. IMEI values can be optionally stored securely in secure IMEI storage **318** of the eUICC **108**. Within individual secure storage containers of the eUICC **108**, e.g., within profile issuer security domains (ISD-Ps) **320**, such as ISD-P-a **320a** and ISD-P-b **320b**, most recently used IMEI values for the profile of that ISD-P **320** can be maintained. For example, ISD-P-a **320a** shows IMEI2 as the most recently used IMEI value for the profile

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of ISD-P-a, and ISD-P-b 320b shows IMEI3 as the most recently used IMEI value for the profile of ISD-P-b. The IMEI binding logic 316 of the baseband module 308 can send an ES10c proprietary command bindIMEIRequest to the eUICC 108 indicating a request to bind a particular SIM/eSIM (identified by an ICCID value) with a particular IMEI value. The eUICC 108 can subsequently provide a response indicating acceptance of the binding. The IMEI binding logic 316 of the baseband module 308 can also query the eUICC 108 for currently used IMEI bindings, such as by sending an ES10c GetProfilesInfo command including a proprietary tag to indicate a request for the IMEI bindings in use by the eUICC 108. The eUICC 108 can respond by sending its current IMEI binding information to the IMEI binding logic 316.

An exemplary sequence of actions to bind an IMEI value to an eSIM 208 can include the following. An eSIM 208 can be installed into an ISD-P (applicable MNO security domain) of the eUICC 108 and not be enabled initially. There can be no IMEI value associated with the newly installed eSIM 208. The eSIM 208 can be subsequently enabled and associated with a cellular wireless service subscription of the mobile wireless device 102. The IMEI binding logic 316 of the baseband module 308 can send the proprietary ES10c command (bindIMEIRequest) to the eUICC 108 including a particular IMEI value and the ICCID value of the eSIM 208 in the proprietary ES10c command. The eUICC 108 can store the particular IMEI value for the eSIM 208 in the ISD-P (MNO security domain). Each eSIM 208 installed on the eUICC 108 can have its own ISD-P (MNO security domain). The baseband module 308, via the IMEI binding logic 316, can subsequently query the eUICC 108 for IMEI binding information using a proprietary tag in a GetProfilesInfo ES10c command. The IMEI binding logic 316 can use a history of associations (bindings) of IMEI values to eSIMs 208 (and to physical SIMs) as an input to determine how to assign IMEI values when installing and/or when enabling eSIMs 208 on the eUICC 108 (or when installing a pSIM in the mobile wireless device 102).

Different MNOs can have different policies regarding the use of IMEI values. Some MNOs keep track of and care about IMEI values associated with cellular wireless subscriptions, while some MNOs do not. Carrier bundle information, provided to the mobile wireless device 102 during manufacture in the factory or afterward via an update in the field, can include information indicating whether an MNO associates IMEI values with cellular wireless subscriptions in their network infrastructure back end systems. When the IMEI binding logic 316 determines that two different cellular wireless subscriptions may be assigned to an identical IMEI value, e.g., when they each have the same most recently used IMEI value and both are to be active at the same time, the IMEI binding logic 316 can use the carrier bundle information to determine whether IMEI bindings matter to the MNOs with which the cellular wireless subscriptions are associated. For those MNOs that do maintain IMEI value associations, the IMEI binding logic 316 can seek to use the same IMEI value for those MNOs, i.e., to use a particular IMEI value consistently for the SIMs/eSIMs of that MNO.

FIG. 4 illustrates a table 400 of exemplary ES10c IMEI binding commands, including the proprietary BindIMEIRequest ES10c command, and an associated proprietary BindIMEIResponse ES10c command. A proprietary tag can also be added to a standard GetProfilesInfo ES10c command (not shown). When the proprietary tag is included in the GetProfilesInfo ES10c command, the eUICC 108 can return

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IMEI binding information for each eSIM 208 installed on the eUICC 108. If no IMEI binding information for an eSIM 208 is available, the eUICC 108 can provide a special value indicating so.

FIG. 5 illustrates a block diagram 500 of an exemplary system for dynamic IMEI provisioning for a mobile wireless device 102. During manufacturing of the mobile wireless device 102, a limited number of IMEI values can be assigned to the mobile wireless device 102, e.g., one IMEI value for a single SIM/eSIM device and two IMEI values for a DSDS (e.g., dual SIM or SIM+eSIM) device. Additional IMEI values can be assigned to the mobile wireless device 102 as needed after manufacture of the device (e.g., in the field when SIMs/eSIMs in active use by the mobile wireless device 102 change). The telephony module 302 of the mobile wireless device 102 can obtain an IMEI value when a trigger condition occurs. Example triggers can include: i) an eligibility check for an eSIM 208 during an eSIM installation procedure, ii) a post eSIM installation procedure to acquire a spare IMEI value in anticipation of a future eSIM installation, and iii) upon device initialization (post-sale unboxing or device reset by a user). The telephony module 302 can establish a secure communication channel with a dynamic IMEI provisioning front end server 504, e.g., via transmission of a CreateSession command and receipt of an applicable response, e.g., a randomly assigned one-time SessionID value. The telephony module 302 can subsequently send a command, e.g., a CheckAvailableOptions command, that includes a value indicating a request to check for IMEI values and/or for acquiring one or more new IMEI values for the mobile wireless device 102. The CheckAvailableOptions command can also include an eUICC signed payload for verification by the dynamic IMEI provisioning front end server 504. The dynamic IMEI provisioning front end server 504 can return one or more IMEI values. The IMEI values returned by the dynamic IMEI provisioning front end server 504 can be signed by the OEM for validation before storing the IMEI values for use on the mobile wireless device 102. In some embodiments, a check for IMEI values returns those IMEI values already assigned to the mobile wireless device 102. In some embodiments, a request for new IMEI values returns one or more newly assigned IMEI values for use by the mobile wireless device 102. In some embodiments, the dynamic IMEI provisioning front end server 504 is managed by an original equipment manufacturer (OEM) of the mobile wireless device 102 and/or by a third party. In some embodiments, an IMEI back end server 506 can provide storage of IMEI values in use by mobile wireless devices 102 of the OEM. The dynamic IMEI provisioning front end server 504 and/or the IMEI back end server 506 can communicate with one or more MNO servers 508, such as when assigning IMEI values to a mobile wireless device 102 or in response to queries from an MNO server 508 regarding IMEI values assigned to the mobile wireless device 102. IMEI values dynamically assigned to the mobile wireless device 102 can be securely stored in the secure IMEI storage 310 of the baseband module 308 of the mobile wireless device 102. In some embodiments, IMEI values for the mobile wireless device 102 can also be stored securely in secure IMEI storage 318 on the eUICC 108. In some embodiments, a particular IMEI value can be bound to a physical SIM, while a set of additional IMEI values can be used for eSIMs 208. In some embodiments, an IMEI value for a physical SIM can be stored in the eUICC 108. Communication of IMEI information, e.g., IMEI binding assignments and IMEI binding information responsive to queries, between the baseband

module **308** and the eUICC **108** can be realized by IMEI assignment logic **502**. In some embodiments, the IMEI binding logic **316** of FIG. **3** and the IMEI assignment logic **502** of FIG. **5** can be realized in a combined logic element of the baseband module **308**.

FIG. **6** illustrates a detailed view of a representative computing device **600** that can be used to implement various methods described herein, according to some embodiments. In particular, the detailed view illustrates various components that can be included in the mobile wireless device **102**. As shown in FIG. **6**, the computing device **600** can include a processor **602** that represents a microprocessor or controller for controlling the overall operation of computing device **600**. The computing device **600** can also include a user input device **608** that allows a user of the computing device **600** to interact with the computing device **600**. For example, the user input device **608** can take a variety of forms, such as a button, keypad, dial, touch screen, audio input interface, visual/image capture input interface, input in the form of sensor data, etc. Still further, the computing device **600** can include a display **610** that can be controlled by the processor **602** to display information to the user. A data bus **616** can facilitate data transfer between at least a storage device **640**, the processor **602**, and a controller **613**. The controller **613** can be used to interface with and control different equipment through an equipment control bus **614**. The computing device **600** can also include a network/bus interface **611** that communicatively couples to a data link **612**. In the case of a wireless connection, the network/bus interface **611** can include a wireless transceiver.

The computing device **600** also includes a storage device **640**, which can comprise a single disk or a plurality of disks (e.g., hard drives), and includes a storage management module that manages one or more partitions within the storage device **640**. In some embodiments, storage device **640** can include flash memory, semiconductor (solid state) memory or the like. The computing device **600** can also include a Random Access Memory (RAM) **620** and a Read-Only Memory (ROM) **622**. The ROM **622** can store programs, utilities or processes to be executed in a non-volatile manner. The RAM **620** can provide volatile data storage, and stores instructions related to the operation of the computing device **600**. The computing device **600** further includes a secure element **624**, which can include an eUICC **108** on which to store one or more eSIMs **208** and/or a UICC **118** (physical SIM card).

Representative Embodiments

In some embodiments, a method for binding IMEI values with physical SIMs included in UICCs **118** of a mobile wireless device **102** and/or with eSIMs **208** included in an eUICC **108** of the mobile wireless device **102** can include: i) determining a configuration of physical SIMs and eSIMs **208** of the mobile wireless device **102**; ii) determining a subscription mapping of the physical SIMs and eSIMs **208** that are in active use, e.g., enabled; iii) determining whether the active SIMs and eSIMs **208** have been previously associated with IMEI values; iv) selecting an IMEI value stored in a secure storage of the mobile wireless device **102** to associate with an eSIM of the mobile wireless device **102**; and v) sending, to the eUICC **108** by a processor **104** of the mobile wireless device **102**, a command to bind the selected IMEI value to the eSIM, where the selected IMEI value is a most recently used IMEI value previously associated with the eSIM.

In some embodiments, the eUICC stores most recently used IMEI values for SIMs and eSIMs **208** of the mobile wireless device **102** in a secure storage container on the eUICC **108** of the mobile wireless device **102**. For example, the secure storage container can be an ISD-P or an MNO security domain (SD) on the eUICC **108**. In some embodiments, the eUICC **108** blocks over-the-air updates of most recently used IMEI values to protect against denial of service (DoS) attacks. In some embodiments, a most recently used IMEI value for an ICCID or a physical SIM can be saved in a secure storage of the mobile wireless device **102**, e.g., on the eUICC **108** and/or in a memory external to the eUICC **108**.

In some embodiments, a mobile wireless device **102** binds IMEI values based on most-recently used IMEI values for one or more physical SIMs and/or eSIMs **208** of the mobile wireless device **102**. In some embodiments, when a physical SIM and/or an eSIM **208** is used for a single SIM subscription or a DSIDS configuration, the mobile wireless device **102** can query secure storage for most-recently used IMEI information. The IMEI information can be stored securely in an eUICC **108** of the mobile wireless device **102** and/or in secure storage of the mobile wireless device **102** that is external to the eUICC **108**. In some embodiments, the mobile wireless device **102** selects a most-recently used IMEI value to associate with a physical SIM or an eSIM **208** to reduce a probability of changing IMEI values for a particular cellular wireless service subscription, which may otherwise trigger a fraud detection mechanism at an MNO infrastructure system or at another information technology (IT) system.

In some embodiments, a mobile wireless device **102** binds IMEI values based on an MNO configuration for both physical SIMs and eSIMs. The mobile wireless device **102** can maintain MNO configurations for each MNO, including whether an MNO provisioning system accounts for IMEI values associated with physical SIMs and/or eSIMs. When binding an IMEI value to a physical SIM or eSIM **208**, the mobile wireless device **102** can prioritize associating an IMEI values based on a past history of IMEI value usage, particularly with cellular wireless service subscriptions for MNOs that account for the IMEI values. In some embodiments, when two cellular wireless service subscriptions have been previously associated with an identical IMEI value, the mobile wireless device **102** prioritizes using the same IMEI value with the MNO that accounts for (has a dependency on) the IMEI value selected for association.

In some embodiments, a mobile wireless device **102** can dynamically provision IMEI values after manufacture. A network-based server can include logic to determine whether to assign one or more new IMEI values to the mobile wireless device **102**. An eUICC **108** of the mobile wireless device **102** can send signed payloads to the network-based server, providing attestation to possession by the mobile wireless device **102** of an active eUICC **108**. The network-based service can use eUICC identifier (EID) information included in a certificate provided by the eUICC **108** to determine whether the eUICC **108** is associated with a particular original equipment manufacturer (OEM). The mobile wireless device **102** can send a list of meta-data to the network-based server to use as an input for IMEI assignment logic executing on the network-based server. Representative meta-data can include a device serial number and information about one or more IMEI values presently stored on the mobile wireless device **102**.

In some embodiments, a mobile wireless device **102** uses one or more triggers to request assignment of an IMEI value

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for one or more SIMs and/or eSIMs 208 of the mobile wireless device 102. Representative triggers can include determining new IMEI values are required during a device initialization procedure, e.g., during an unboxing or after a factory reset of the mobile wireless device 102. In some embodiments, the mobile wireless device 102 can determine to obtain a new IMEI value after installation of an eSIM 208 to an eUICC 108 of the mobile wireless device 102, where the new IMEI value can be stored in preparation for future installation of another eSIM 208 to the eUICC 108 of the mobile wireless device 102. The mobile wireless device 102 can assign the stored IMEI value during installation of the another eSIM 208. In some embodiments, an eSIM installation procedure can include a handshake process with a network-based server that manages assignment of IMEI values for eSIMs under installation.

In some embodiments, the mobile wireless device 102 securely stores IMEI values in an eUICC 108 of the mobile wireless device 102. IMEI values can be signed by a known entity for which a root key is pre-loaded on the eUICC 108 to establish a trust relationship between the known entity and the eUICC 108. The eUICC 108 of the mobile wireless device 102 can validate signatures that accompany IMEI values provided by the known entity before storing the IMEI values. The eUICC 108 of the mobile wireless device 102 can prevent unauthorized updates of stored IMEI values, including blocking changes to IMEI values by SIM/eSIM OTA updates. In some embodiments, the eUICC 108 stores IMEI values in a secure domain, e.g., an ISD-R, and prevents unauthorized access to the stored IMEI values, such as disallowing applets stored in an eSIM 208 from accessing or changing the stored IMEI values.

In some embodiments, a mobile wireless device 102 dynamically informs one or more MNO servers of IMEI values assigned to SIMs/eSIMs. The MNO servers can be apprised of the IMEI value assignments after the mobile wireless device 102 has been manufactured, i.e., post-factory. In some embodiments, the mobile wireless device 102 creates application programming interfaces (APIs) and notifications to inform MNOs of assigned IMEI values. In some embodiments, IMEI information for one or more SIMs and/or one or more eSIMs of the mobile wireless device 102 are aggregated to include both IMEI values assigned during manufacture and IMEI values dynamically assigned post-manufacture.

Wireless Terminology

In accordance with various embodiments described herein, the terms “wireless communication device,” “wireless device,” “mobile wireless device,” “mobile station,” and “user equipment” (UE) may be used interchangeably herein to describe one or more common consumer electronic devices that may be capable of performing procedures associated with various embodiments of the disclosure. In accordance with various implementations, any one of these consumer electronic devices may relate to: a cellular phone or a smart phone, a tablet computer, a laptop computer, a notebook computer, a personal computer, a netbook computer, a media player device, an electronic book device, a MiFi® device, a wearable computing device, as well as any other type of electronic computing device having wireless communication capability that can include communication via one or more wireless communication protocols such as used for communication on: a wireless wide area network (WWAN), a wireless metro area network (WMAN), a wireless local area network (WLAN), a wireless personal area network (WPAN), a near field communication (NFC), a cellular wireless network, a fourth generation (4G) Long

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Term Evolution (LTE), LTE Advanced (LTE-A), and/or 5G or other present or future developed advanced cellular wireless networks.

The wireless communication device, in some embodiments, can also operate as part of a wireless communication system, which can include a set of client devices, which can also be referred to as stations, client wireless devices, or client wireless communication devices, interconnected to an access point (AP), e.g., as part of a WLAN, and/or to each other, e.g., as part of a WPAN and/or an “ad hoc” wireless network. In some embodiments, the client device can be any wireless communication device that is capable of communicating via a WLAN technology, e.g., in accordance with a wireless local area network communication protocol. In some embodiments, the WLAN technology can include a Wi-Fi (or more generically a WLAN) wireless communication subsystem or radio, the Wi-Fi radio can implement an Institute of Electrical and Electronics Engineers (IEEE) 802.11 technology, such as one or more of: IEEE 802.11a; IEEE 802.11b; IEEE 802.11g; IEEE 802.11-2007; IEEE 802.11n; IEEE 802.11-2012; IEEE 802.11ac; or other present or future developed IEEE 802.11 technologies.

Additionally, it should be understood that the UEs described herein may be configured as multi-mode wireless communication devices that are also capable of communicating via different third generation (3G) and/or second generation (2G) RATs. In these scenarios, a multi-mode UE can be configured to prefer attachment to LTE networks offering faster data rate throughput, as compared to other 3G legacy networks offering lower data rate throughputs. For instance, in some implementations, a multi-mode UE may be configured to fall back to a 3G legacy network, e.g., an Evolved High Speed Packet Access (HSPA+) network or a Code Division Multiple Access (CDMA) 2000 Evolution-Data Only (EV-DO) network, when LTE and LTE-A networks are otherwise unavailable.

The various aspects, embodiments, implementations or features of the described embodiments can be used separately or in any combination. Various aspects of the described embodiments can be implemented by software, hardware or a combination of hardware and software. The described embodiments can also be embodied as computer readable code on a non-transitory computer readable medium. The non-transitory computer readable medium is any data storage device that can store data which can thereafter be read by a computer system. Examples of the non-transitory computer readable medium include read-only memory, random-access memory, CD-ROMs, HDDs, DVDs, magnetic tape, and optical data storage devices. The non-transitory computer readable medium can also be distributed over network-coupled computer systems so that the computer readable code is stored and executed in a distributed fashion.

Regarding the present disclosure, it is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the described embodiments. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the described embodiments.

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Thus, the foregoing descriptions of specific embodiments are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the described embodiments to the precise forms disclosed. It will be apparent to one of ordinary skill in the art that many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. A method for binding international mobile equipment identifier (IMEI) values with electronic subscriber identity modules (eSIMs) on a wireless device, the method comprising:

by a processor of the wireless device external to an embedded universal integrated circuit card (eUICC) of the wireless device;

detecting a first command to enable a first eSIM on the eUICC of the wireless device;

determining whether the first eSIM has been previously associated with an IMEI value based on IMEI binding information;

when the first eSIM has not been previously associated with an IMEI value, obtaining from a network-based server an assigned IMEI value for the first eSIM;

when the first eSIM has been previously associated with an IMEI value, selecting a most recently used IMEI value for the first eSIM; and

sending a second command to the eUICC to bind the assigned IMEI value or the most recently used IMEI value to the first eSIM,

wherein the second command comprises an ES10c.bindIMEI command to bind the assigned IMEI value or the most recently used IMEI value with the first eSIM.

2. The method of claim 1, wherein the IMEI binding information is obtained via a query command sent to the eUICC.

3. The method of claim 1, wherein the IMEI binding information indicates one or more associations between IMEI values and international circuit card identifier (ICCID) values of one or more eSIMs on the eUICC.

4. The method of claim 1, wherein the IMEI binding information indicates an association between an international circuit card identifier (ICCID) value of the first eSIM and the most recently used IMEI value for the first eSIM.

5. The method of claim 1, wherein the IMEI binding information indicates no IMEI binding information for the first eSIM is available.

6. The method of claim 1, wherein the proprietary ES10c command includes i) the assigned IMEI value or the most recently used IMEI value and ii) an integrated circuit card identifier (ICCID) value of the first eSIM.

7. The method of claim 1, further comprising:

by the processor of the wireless device external to the eUICC:

detecting a third command to enable a second eSIM of the eUICC of the wireless device;

determining the second eSIM of the eUICC has been previously associated, most recently, with the most recently used IMEI value for the first eSIM; and

when respective cellular wireless subscriptions for the first eSIM and the second eSIM allow identical IMEI values to be used for multiple active eSIMs on the eUICC, sending a command to the eUICC to bind the most recently used IMEI value for the first eSIM to the second eSIM.

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8. The method of claim 1, further comprising:

by the processor of the wireless device external to the eUICC:

selecting IMEI values to bind to one or more eSIMs when installing and/or when enabling the one or more eSIMs on the eUICC based on a history of bindings of IMEI values to eSIMs on the eUICC of the wireless device.

9. The method of claim 1, further comprising:

by the eUICC of the wireless device:

receiving an over-the-air (OTA) update for the first eSIM from a mobile network operator (MNO) server associated with the first eSIM; and

blocking changes by the OTA update to the IMEI value bound to the first eSIM.

10. The method of claim 1, further comprising:

by the eUICC of the wireless device:

maintaining, within one or more mobile network operator (MNO) security domains on the eUICC, records of most recently used IMEI values bound to eSIMs in the MNO security domains.

11. An apparatus configured for binding international mobile equipment identifier (IMEI) values with electronic subscriber identity modules (eSIMs) on a wireless device, the apparatus comprising:

a processor of the wireless device external to an embedded universal integrated circuit card (eUICC) of the wireless device, the processor communicatively coupled to a memory storing instructions that, when executed by the processor, cause the processor to perform actions that include:

detecting a first command to enable a first eSIM on the eUICC of the wireless device;

determining whether the first eSIM has been previously associated with an IMEI value based on IMEI binding information;

when the first eSIM has not been previously associated with an IMEI value, obtaining from a network-based server an assigned IMEI value for the first eSIM;

when the first eSIM has been previously associated with an IMEI value, selecting a most recently used IMEI value for the first eSIM; and

sending a second command to the eUICC to bind the assigned IMEI value or the most recently used IMEI value to the first eSIM,

wherein the second command comprises an ES10c.bindIMEI command to bind the assigned IMEI value or the most recently used IMEI value with the first eSIM.

12. The apparatus of claim 11, wherein the IMEI binding information is obtained via a query command sent to the eUICC.

13. The apparatus of claim 11, wherein the IMEI binding information indicates one or more associations between IMEI values and ICCID values of one or more eSIMs on the eUICC.

14. The apparatus of claim 11, wherein the IMEI binding information indicates an association between an international circuit card identifier (ICCID) value of the first eSIM and the most recently used IMEI value for the first eSIM.

15. The apparatus of claim 11, wherein the IMEI binding information indicates no IMEI binding information for the first eSIM is available.

16. The apparatus of claim 11, wherein the proprietary ES10c command includes i) the assigned IMEI value or the most recently used IMEI value and ii) an integrated circuit card identifier (ICCID) value of the first eSIM.

17. A non-transitory computer-readable medium storing instructions for binding international mobile equipment

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identifier (IMEI) values with electronic subscriber identity modules (eSIMs) on a wireless device, where the instructions, when executed on a processor of the wireless device external to an embedded universal integrated circuit card (eUICC) of the wireless device, configure the processor to 5 perform actions including:

- detecting a first command to enable a first eSIM on the eUICC of the wireless device;
- determining whether the first eSIM has been previously associated with an IMEI value based on IMEI binding 10 information;
- when the first eSIM has not been previously associated with an IMEI value, obtaining from a network-based server an assigned IMEI value for the first eSIM;
- when the first eSIM has been previously associated with 15 an IMEI value, selecting a most recently used IMEI value for the first eSIM; and
- sending a second command to the eUICC to bind the assigned IMEI value or the most recently used IMEI value to the first eSIM, 20

wherein the second command comprises an ES10c.bindIMEI command to bind the assigned IMEI value or the most recently used IMEI value with the first eSIM.

18. The non-transitory computer-readable medium of 25 claim **17**, wherein the IMEI binding information is obtained via a query command sent to the eUICC.

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